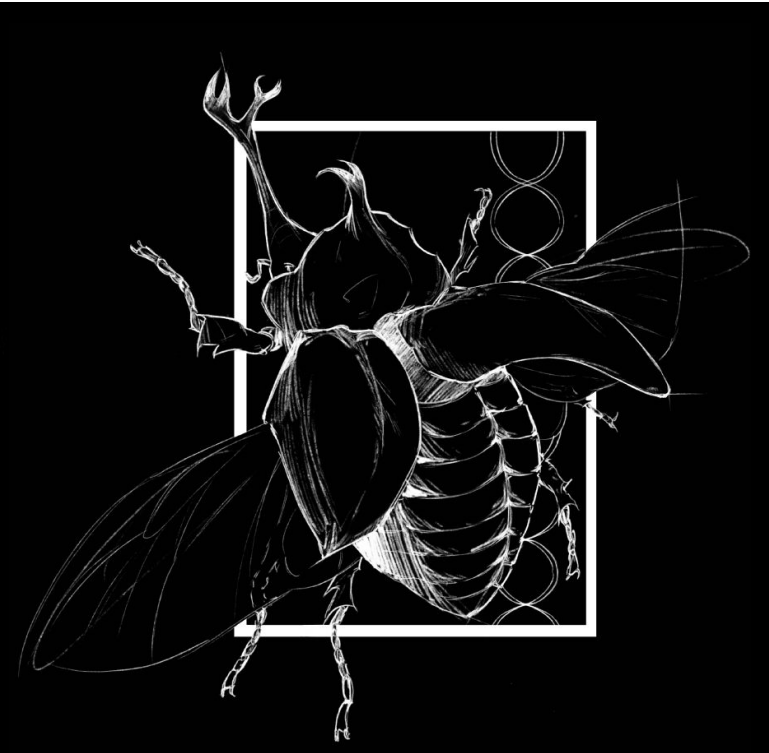


Blackmon Lab

Evolutionary Biology



The Rate of Chromosome Evolution and Genome Size in Clade *Liliaceae*

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Biology

Background

Genome size varies drastically across various flowering species across the plant kingdom. With such diversity, fundamental questions regarding the size of a plant's genome, including its causes and effects, arise. In particular, it becomes imperative to question if this change in size bears any impact on chromosomal dynamics and counts. The clade *Liliaceae* represents a good basis for examining this relationship. There is a notable amount of variation in regards to genome size across the clade that also corresponds to the high degree of conservation of morphological characteristics.

The clade *Liliaceae* consists of approximately 15-16 genera and approximately 600 known species that are predominantly distributed across the Northern Hemisphere of the world. Plants within this clade have a great amount of diversity in genome size, varying from small genome sizes to some of the biggest sizes recorded among flowering plants (angiosperms). The *Fritillaria* species have been known to have a large genome at 85 Gbp. The karyotypic diversity of clade has also been well documented, which allows *Liliaceae* to serve as a great model to study the relationship between these two traits.

Chromosome evolution can be described as any changes to chromosome number, structure, and organization over time. Mechanisms of change for these include polyploidy or chromosomal rearrangements like inversions, duplications, and translocations. Predictions based on studies in other clades may suggest that larger genomes may experience different rates of chromosomal changes compared to smaller genomes. Understanding this relationship can help people understand if chromosomal changes influence speciation rates, adaptation, and reproductive isolation.

This study aims to investigate the relationship between genome size and the rate of chromosome evolution across clade *Liliaceae* using comparative phylogenetic methods and molecular phylogenetic data. The hypothesis that is being tested within this study is that larger genomes will exhibit increased rates of chromosomal change. The goal of this study is to have findings that contribute to understanding how evolutionary forces can shape plant genomes and karyotypes.

Study Design & AI

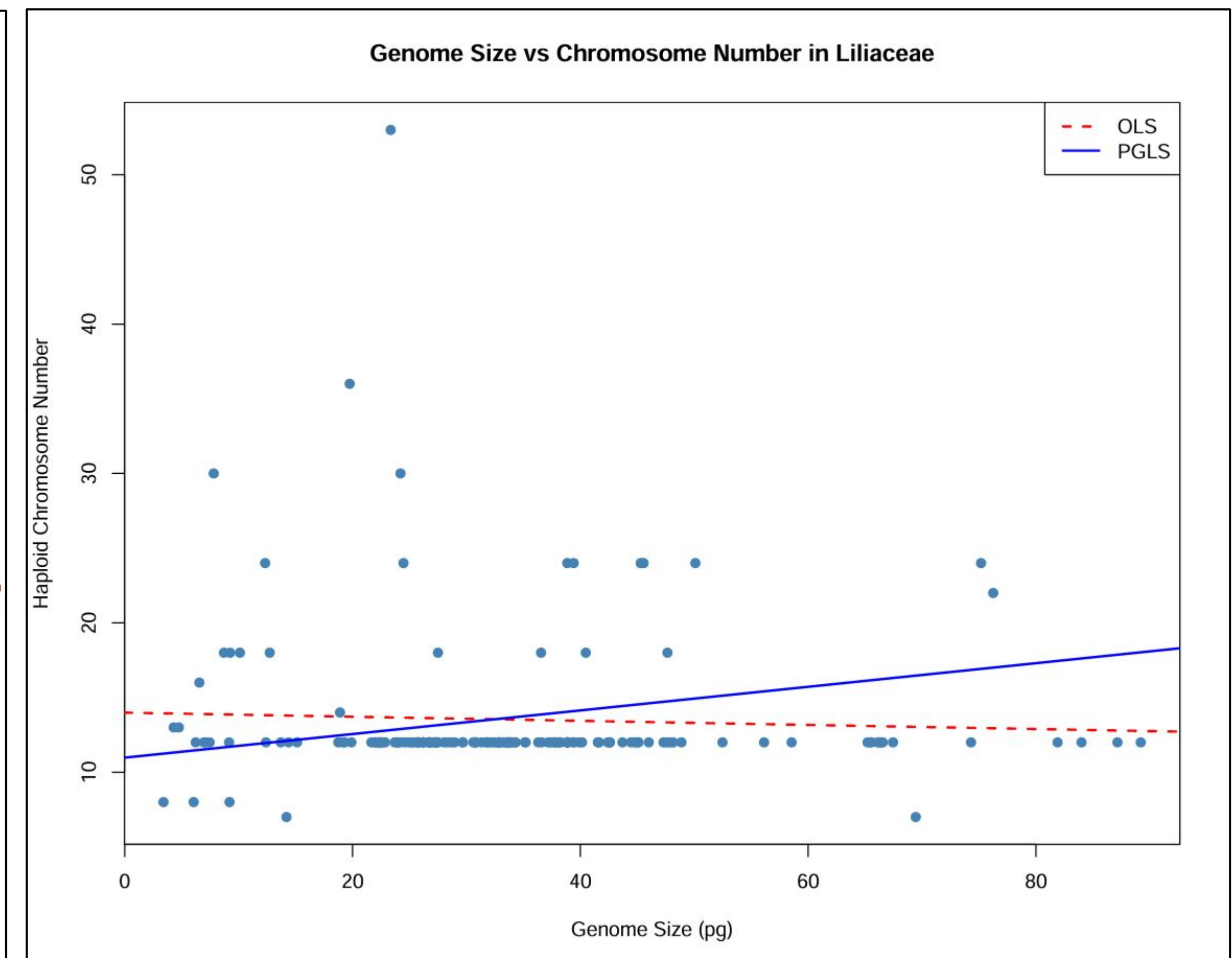
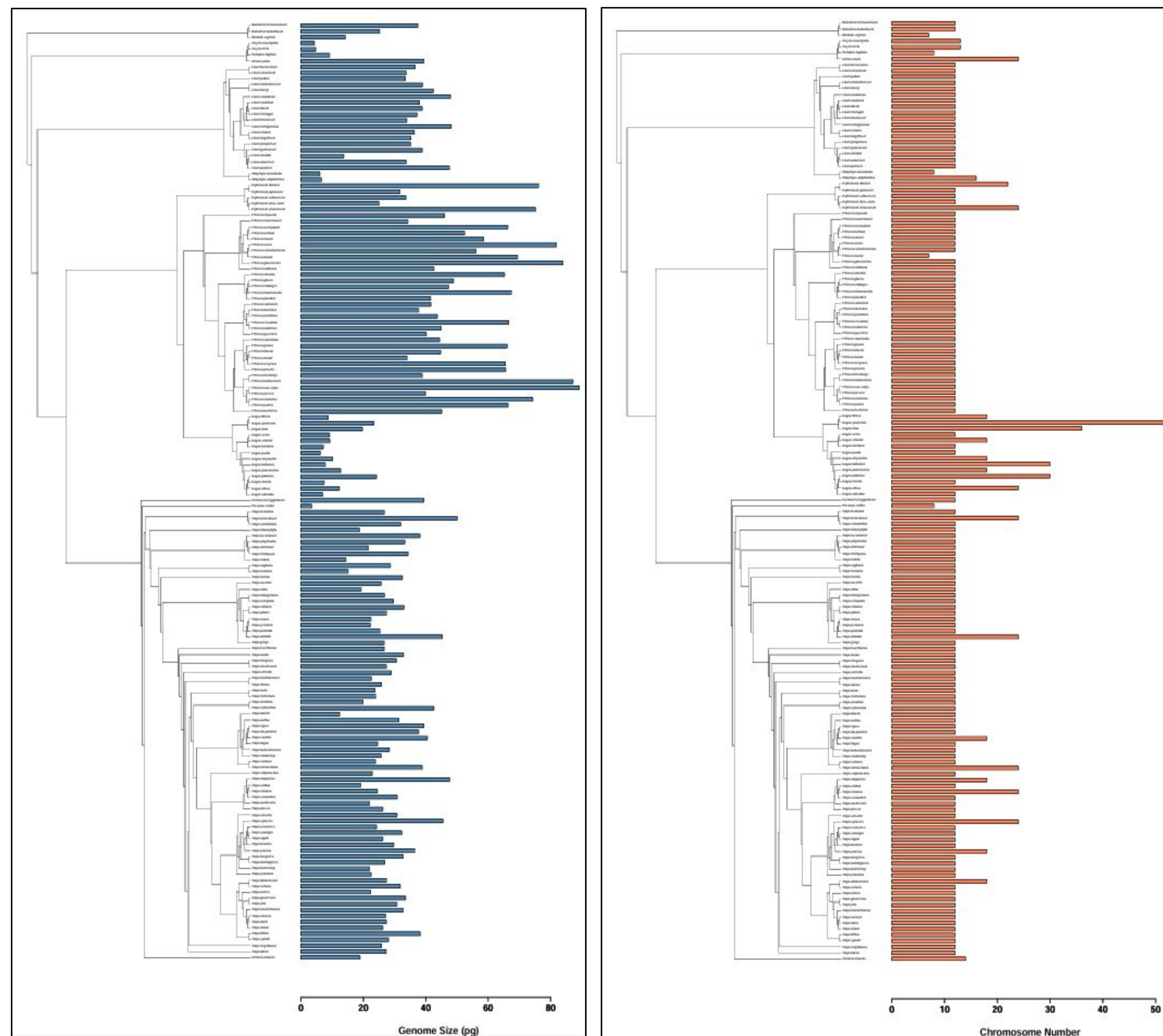
This study employed a phylogenetic comparative approach to investigate the relationship between genome size and chromosome number across species within the clade *Liliaceae*. A molecular phylogeny was reconstructed or obtained from published sources to establish evolutionary relationships among the sampled taxa. To account for the non-independence of species data due to shared evolutionary history, a Phylogenetic Generalized Least Squares (PGLS) regression analysis model was assumed. The PGLS model utilized a Brownian motion (BM) model of trait evolution to help fit the traits. This model assumes that traits evolve randomly along phylogenetic branches with variance proportional to branch length. It also assumes that trait changes occur gradually yet randomly over evolutionary time, making it appropriate for continuous characters such as genome size and chromosome number.

All the data mapping and analysis was conducted and visualized in RStudio. The phylogeny was sourced from Smith and Brown, 2018. The data for the chromosome number (haploid number) was sourced from Rice et. al, 2015 and the data from the genome size in picograms (pg) was sourced from Leitch et. al, 2019. The data of the chromosome number and genome size were listed together and were matched based on the availability. The PGLS analyses were conducted using the caper and/or nlme packages, with phylogenetic trees handled via the ape package. Visualizations, including phylogenetic plots and regression figures, were generated using ggplot2 and ggtree packages. Claude Opus 4.5 was utilized for R script generation and optimization to help with the PGLS analysis and visualization. All AI-generated scripts were adequately reviewed and modified as needed.

Model Fitting

Data matches for 158 species across 14 genera were successfully matched. These 158 species were then mapped to the phylogenetic tree to create a pruned mapped tree showing genome size and chromosome number. All 158 species were mapped onto the original phylogenetic tree. The genome size of the 158 selected species ranged from 3.4 pg (*Prosartes smithii*) to 89.2 pg (*Fritillaria uva-vulpis*), while the chromosome number ranged from a haploid number of 7 (*Fritillaria davisii*, *Medeola virginica*) to a haploid number of 53 (*Gagea spathacea*). As both traits were continuous and ranged upon a spectrum, the Phylogenetic Generalized Least Squares (PGLS) model was selected for the data analysis model after the trait data was collected and mapped. The first analysis ran aimed to look at the phylogenetic signal for the two traits, which is the tendency for closely related species to have similar trait values. For each trait, the following values were calculated: Blomberg's K (Brownian motion expectation), Pagel's λ (evolution of trait in relation to phylogeny), and their respective p-values. Finally, the PGLS model results were computed and plotted, along with an Ordinary Least Squares model for comparison. The λ , slope, R², and the p-value of λ were computed.

Results



For chromosome number, the Blomberg K value was found to be 0.022 with a p-value 0.636 and Pagel's λ of 0.346 with a p-value of 0.001. For genome size, the Blomberg K value was found to be 0.068 with a p-value 0.001 and Pagel's λ of 0.691 with a p-value of <0.001. Given these results it was concluded that chromosome number showed weak to moderate phylogenetic signal, meaning there was some phylogenetic constraint but was still relatively independent across *Liliaceae* lineages. For genome size, it was concluded that there was moderate to strong phylogenetic signal and there was a notable amount of phylogenetic constraint in how genome size evolved across *Liliaceae* lineages as closely related species had similar genome sizes.

For the PGLS model, the slope was calculated to be 0.079 with a p-value of 0.0059 and an R² of 0.048. The λ value for the PGLS model was found to be 0.505. Based on these values, there is an overall positive and statistically significant (p-value < 0.01) correlation between genome size and chromosome number. However, there is only a small effect size as the R² indicates that 4.8% of the variation in chromosome number is explained by genome size. The moderate λ implies that both traits, when examined together, are partially influenced by phylogeny.

References

Leitch IJ, Johnston E, Pellicer J, Hidalgo O, Bennett MD. 2019. Angiosperm DNA C-values database (release 9.0, Apr 2019) <https://cvalues.science.kew.org/>

Smith, Stephen A., and Joseph W. Brown. 2018. "Constructing a Broadly Inclusive Seed Plant Phylogeny." American Journal of Botany 105 (3): 302-14.

Rice, Anna, Lior Glick, Shiran Abadi, Moshe Einhorn, Naama M. Kopelman, Ayelet Salman-Minkov, Jonathan Mayzel, Ofer Chay, and Itay Mayrose. 2015. "The Chromosome Counts Database (CCDB) - a Community Resource of Plant Chromosome Numbers." The New Phytologist 206 (1): 19-26.



Discussion

There is only a weak and positive but statistically significant correlation between genome size and chromosome number after controlling for phylogeny using the PGLS model under a Brownian motion of evolution assumption. While phylogeny does influence the correlation between the two traits and suggests that there is an evolutionary relationship between the two, it is not entirely driven by evolutionary relationships, suggesting that there are probably other influences that determine the relationship between chromosome number and genome size. The λ for the PGLS model being approximately half indicates that half of the covariance in the relationship between genome size and chromosome size can be explained by evolutionary history, while the other half can be attributed to factors such as adaptive evolution, measurement error, or other non-phylogenetic factors.

When examining for phylogenetic signals and how well each trait corresponds with the phylogeny on its own, genome size seems more conserved with phylogenetic patterns than chromosome number. This could be reflective of natural processes that shape genome size like gene deletions, transposable element proliferation, gene duplications that are expected gradually over evolutionary time. On the contrary, the weaker phylogenetic signal for chromosome number could be due to more punctuated changes such as polyploidy, aneuploidy, chromosomal fission/fusion.

With this being considered, a possible explanation for the weak correlation between the two traits can be polyploidy that arose from gene duplication events across various genera over time, as several species show signs of this duplication event in the data. Different species could also hold different amounts of genetic information per chromosome, altering the relationship between genome size and chromosome number as genome size can be assumed to be independent of chromosome number. Lastly, chromosomal fusion or fission can change the number of chromosomes but keep the genetic content the same, also establishing independence between the two traits.

Conclusions & Acknowledgements

This study found a weak but statistically significant positive correlation between genome size and chromosome number in *Liliaceae* after controlling for phylogenetic relationships using PGLS. Genome size exhibits stronger phylogenetic conservatism than chromosome number, suggesting that karyotypic evolution is more volatile. While there appears to be a relationship between the two traits, there seems to be external factors influencing the rate of change within these traits. Future studies should aim to use molecular phylogenies and better branch length accuracies and take repetitive DNA sequences into account in order to further understand the evolution of genome size and diversity within *Liliaceae*.

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